EDGEGUARD V1

Technical Project Record — Intelligent Edge Monitoring & Protection System

Team: InnoVolt | Domain: Smart Infrastructure and Industrial Automation | NAVYANTRA 24-hour Hardware Hackathon, KSSEM, Bengaluru

1. System

EDGEGUARD V1 monitors a DC motor using LM35 temperature, INA219 voltage/current and MPU6050 vibration sensing. An ESP32 performs local risk assessment and protection. A web dashboard displays telemetry and a Python service provides Isolation Forest anomaly detection.

2. Risk Engine

R = 0.4 RT + 0.3 RC + 0.3 RV. NORMAL is below 40, WARNING is 40 to below 70, and CRITICAL is 70 or above. These values are prototype design choices.

3. Protection

At a critical condition the ESP32 can disable the motor path, trip the relay and activate local warning indicators. The ML layer does not directly control the relay.

4. ML

The documented model uses StandardScaler followed by IsolationForest with 200 estimators, contamination 0.05 and random_state 42. Features are temperature, voltage, current and vibration.

5. EMT Relevance

The project involves electromagnetic motor force, back EMF, inductive switching, current sensing, electromagnetic interference, grounding and wireless communication.

6. Data / Validation

The documented collection run contained 257 samples. The available record does not establish that every sample was independently verified as a clean normal running-motor sample. No independent validation accuracy or verified false-alarm percentage is claimed.

7. Limitations

Further work should use independent time-separated data, labelled fault experiments, measured false-alarm rates, improved vibration features and validated health/RUL models if required.